Vibration-Assisted Smart Alginate Mixing Spatula

Madhavkrishnan K S
Dept of Electronics and Communication Engineering
SRM Institute of Science and Technology, Ramapuram

Nishanthan R
Dept of Electrical and Electronics Engineering
Easwari Engineering College, Ramapuram

Abstract

Accurate mixing of alginate is critical in dental procedures to ensure precise impressions. Conventional manual mixing relies heavily on operator experience, leading to inconsistencies in viscosity and quality.

This project proposes a **smart vibratory spatula** that assists mixing while simultaneously providing **real-time feedback on material consistency**. The system utilizes a **motor load-based sensing mechanism**, where variations in motor current are correlated with changes in alginate viscosity.

A microcontroller processes the sensed data and indicates the mixing state using **LED indicators**, enabling the user to identify the optimal consistency without guesswork. The system is compact, hygienic, and cost-effective, making it suitable for practical dental applications.

I. OBJECTIVE

- To design a **handheld device** that improves alginate mixing quality
- To reduce **human error and variability** in mixing
- To implement a **non-contact consistency sensing method**
- To provide **real-time visual feedback** to the user
- To develop a **low-cost, portable, and hygienic solution**.

II. PROBLEM STATEMENT

Manual mixing of alginate suffers from:

- Inconsistent viscosity due to human variation
- Formation of air bubbles and lumps
- Lack of objective indication of optimal mixing stage
- Dependence on operator experience

This results in **poor dental impressions and material wastage**.

III. PROPOSED SOLUTION

The proposed system is a **vibration-assisted smart spatula** that:

- Enhances mixing using a **vibration motor**
- Monitors consistency using **motor current sensing**
- Provides feedback via **LED indicators**

The system eliminates guesswork by informing the user when the mixture reaches **optimal consistency**.

CIRCUIT DIAGRAM

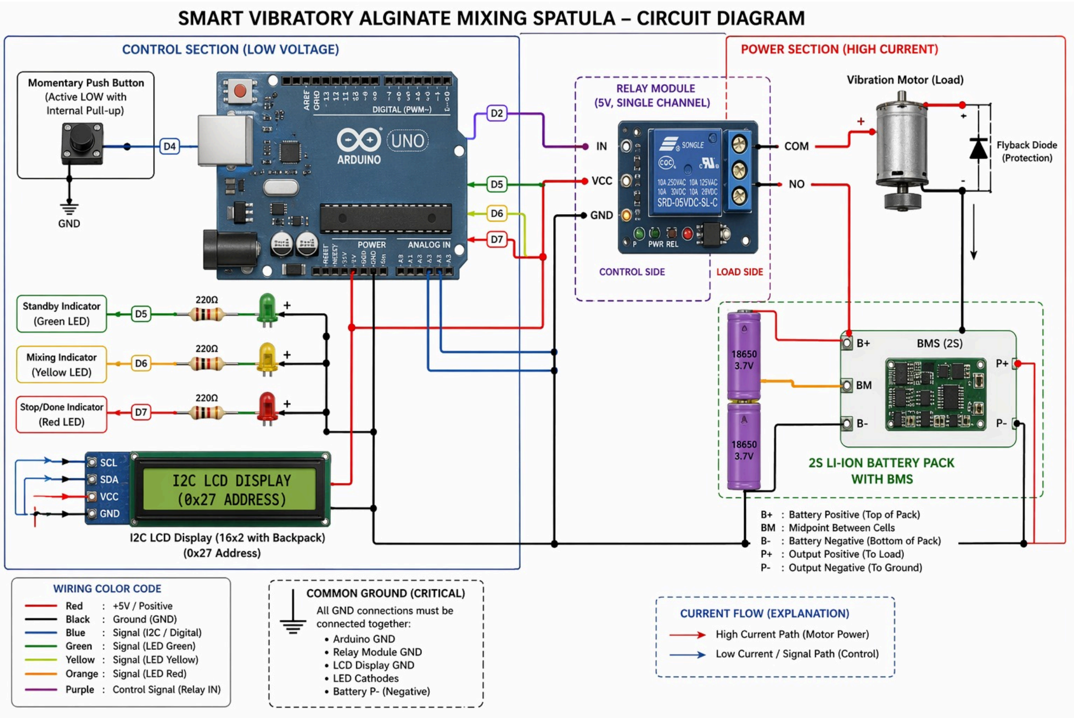

Fig. 1: Complete Circuit Diagram of the Proposed System

IV. WORKING PRINICIPLE

The system operates based on the relationship:

Viscosity ↑ → Resistance ↑ → Motor Load ↑ → Current ↑

As the alginate mixture thickens:

- Resistance to blade motion increases
- The motor requires more torque
- Current drawn by the motor increases

By measuring this current, the system indirectly determines the **consistency of the mixture**

V. SYSTEM ARCHITECTURE

The system consists of three main subsystems:

A. Mechanical System

- Spatula blade
- Handle enclosure
- Eccentric vibration motor

Function: Enhances mixing efficiency and transmits vibration to the material.

B. Electrical System

- 3.7V Li-ion battery
- Power switch

- Charging module
- Wiring and insulation

Function: Supplies power to all components.

C. Sensing and Control System

- Current sensing module (e.g., ACS712 / shunt resistor)
- Microcontroller (Arduino Nano)
- Signal processing logic

Function: Measures motor load and determines consistency.

D. Output System

- Red LED (non-optimal state)
- Green LED (optimal consistency)
- (Optional) Buzzer

Function: Provides user feedback.

VI. DETAILED WORKING

Step 1: Initial State

- Device is OFF
- Alginate powder and water are placed in a bowl
- Spatula is inserted

Default indication: 🔴 Red LED

Step 2: Activation

- User switches ON the device
- Battery powers:
  - Vibration motor
  - Sensing module
  - Microcontroller

Step 3: Mixing Phase

- Motor generates vibration using an eccentric mass
- Vibrations are transmitted to the spatula blade
- User performs manual mixing

Result:

- Faster homogenization
- Reduced lumps and air bubbles

Step 4: Consistency Sensing

- Motor current is continuously measured
- As viscosity changes, current varies accordingly

Conditions:

State	Motor Current	Interpretation
Low	Low	Under-mixed
Medium	Stable range	Optimal
High	High	Over-mixed

Step 5: Signal Processing

To improve reliability:

- Multiple readings are averaged (noise reduction)
- Stability is checked over time (2–3 seconds)

Step 6: Output Indication

- 🔴 Red LED → Non-optimal consistency
- 🟢 Green LED → Optimal consistency

User stops mixing when Green LED appears.

VII. COMPONENTS REQUIRED

Mechanical

- Spatula blade (stainless steel / food-grade polymer)
- Handle enclosure
- Motor mount

Electrical

- DC vibration motor (3–7V)
- Li-ion battery (3.7V)
- Charging module (Type-C / Micro USB)
- Switch

Sensing & Control

- Current sensor (ACS712 recommended)
- Arduino Nano

Output

- Red LED
- Green LED
- Resistors

Miscellaneous

- Connecting wires

- Heat shrink / insulation
- Adhesive / epoxy

VIII. ADVANTAGES :

1. Real-time consistency feedback
2. Non-contact sensing (hygienic)
3. Reduced operator dependency
4. Improved mixing quality
5. Portable and low-cost
6. Easy to use

XI. LIMITATIONS :

1. Sensitive to user mixing force variations
2. Requires calibration for different materials
3. Not a precision viscosity measurement device

XII. FUTURE SCOPE

1. Automatic vibration control (PWM)
2. Buzzer/audio feedback
3. Calibration modes for different materials
4. Bluetooth data logging
5. Replaceable/disposable spatula tips

XIII. CONCLUSION

1. The proposed system successfully introduces a **smart, low-cost, and practical solution** for improving alginate mixing. By utilizing **motor load-based indirect sensing**, the device provides reliable real-time feedback without direct contact with the material.
2. This innovation reduces human error, improves consistency, and enhances the overall quality of dental impressions, making it a valuable tool for clinical applications.